

HyperTrack: Hyperhidrosis Monitoring and Management System

Project Proposal for UCS503

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1 Introduction / Problem Statement

Hyperhidrosis is a condition characterized by excessive sweating that can occur beyond what is necessary for normal temperature regulation. It may affect areas such as the palms, feet, underarms, and face, and can interfere with everyday activities, academic work, professional tasks, and social interactions.

A major practical difficulty for individuals experiencing excessive sweating is the lack of a structured way to monitor their symptoms over time. Users may remember individual episodes but often do not maintain consistent records of the severity, duration, affected body area, possible triggers, activity, or stress level associated with each episode. As a result, it can be difficult to understand whether symptoms are becoming more or less frequent and which circumstances are commonly associated with episodes.

The proposed project, **HyperTrack**, aims to address this problem through a software-based monitoring and management system. The system will allow users to record individual sweating episodes and store the information in a structured database. The collected data will then be processed to generate statistics, visual trends, and personalized observations.

The system will not attempt to medically diagnose hyperhidrosis. Instead, it will function as a monitoring and decision-support tool that helps users organize their own recorded information. A report generation feature will also allow users to summarize their history in a structured format that can be discussed with a healthcare professional.

From a Software Engineering perspective, the project provides a complete software development problem involving requirements analysis, system architecture, database design, user interface development, backend services, data analytics, testing, and future scalability. As an AIML project, the system can additionally incorporate machine learning for personalized pattern analysis and sweating-severity prediction.

2 Objectives

The primary objective of this project is to develop a reliable and user-friendly software system for recording, monitoring, and analyzing hyperhidrosis-related sweating episodes.

The specific objectives are:

- To develop a user-friendly interface for recording sweating episodes and their associated information.
- To store symptom records in a structured and secure database.
- To analyze sweating frequency, severity, duration, affected body areas, and possible triggers.
- To provide dashboards containing graphs, statistics, and historical trends.
- To allow users to maintain treatment or session records without providing medical prescriptions.
- To generate an organized report summarizing the user's recorded history.
- To investigate the use of machine learning for personalized sweating-severity prediction and pattern identification.
- To apply Software Engineering principles including requirements analysis, modular architecture, database design, testing, and iterative development.

3 Proposed Solution

We propose a web-based application named **HyperTrack: Hyperhidrosis Monitoring and Management System**.

The system will consist of the following major components:

- **User Management:** Registration, authentication, and profile management.
- **Sweating Episode Recording:** Users can record severity, duration, affected body area, activity, stress level, and possible triggers.
- **Data Storage:** All recorded information will be stored in a structured relational database.
- **Analytics Module:** The system will calculate frequency, average severity, duration, trigger patterns, and body-area statistics.
- **Dashboard:** Users will be provided with interactive charts and summaries to understand their historical data.
- **Treatment Tracking:** Users can record treatment or session information they are already following.
- **Report Generation:** The application will generate a summarized report containing relevant historical information.
- **Machine Learning Module:** An optional ML component will analyze historical records and predict the likely severity category of future recorded episodes.

4 System Workflow

The core workflow of HyperTrack will follow these stages:

1. The user logs into the application.
2. The user records a sweating episode.

3. The system validates the entered information.
4. The episode is stored in the database.
5. Historical records are processed by the analytics module.
6. The dashboard displays frequency, severity, and trigger trends.
7. The optional machine learning module analyzes historical patterns.
8. The user can generate a structured monitoring report.

The overall data flow can be represented as:

User Input → Episode Recording → Database → Data Analysis → Dashboard → Report

5 Methodology

The methodology of HyperTrack will focus on software engineering, system development, data analysis, and testing. The project will be implemented iteratively so that the core monitoring functionality is completed before advanced analytics and machine learning features are introduced.

5.1 Requirements Analysis

The first phase will identify the functional and non-functional requirements of the system.

Functional requirements will include:

- User registration and authentication.
- User profile management.
- Recording sweating episodes.
- Viewing historical records.
- Dashboard generation.
- Trigger and severity analysis.
- Treatment/session tracking.
- Report generation.

Non-functional requirements will include:

- Usability.
- Reliability.
- Data privacy.
- Maintainability.
- Scalability.
- Acceptable response time.

5.2 System Architecture

The application will use a three-tier architecture:

1. **Presentation Layer:** HTML, CSS, and JavaScript will provide the user interface.
2. **Application Layer:** Python with Flask will provide backend services and REST APIs.
3. **Data Layer:** MySQL will store users, episodes, triggers, treatment records, and analytical information.

This separation will make the system easier to test, maintain, and extend.

5.3 Episode Data Collection

Each sweating episode will contain structured information such as:

- Date and time.
- Sweating severity.
- Duration.
- Affected body area.
- Activity.
- Self-reported stress level.
- Possible trigger.
- Temperature and humidity, where available.
- Additional user notes.

The collected information will form the basis for both statistical analysis and future machine learning functionality.

5.4 Data Processing and Analytics

The stored records will be cleaned and processed before analysis. The system will calculate:

- Daily and weekly episode frequency.
- Average sweating severity.
- Average episode duration.
- Most frequently affected body area.
- Frequently recorded possible triggers.
- Severity trends over time.
- Associations between recorded stress levels and severity.

The results will be displayed using charts and summary statistics.

The system will describe these results as patterns or associations in the user's recorded data rather than medical conclusions.

5.5 Machine Learning Component

As an AIML component, the project will investigate a supervised machine learning model for sweating-severity classification.

Potential input features include:

- Stress level.
- Temperature.
- Humidity.
- Activity level.
- Time of day.
- Previous severity.
- Duration.
- Caffeine intake.
- Affected body area.

The target output will be a severity category:

Low / Moderate / High

Candidate algorithms include Logistic Regression, Decision Trees, and Random Forest. Random Forest will be considered as the initial model because it can handle multiple input features and provide feature importance information.

The ML component will be treated as a supplementary analytical feature and will not be presented as a medical diagnostic system.

5.6 Testing and Validation

Testing will be performed throughout the development process.

- **Unit Testing:** Individual functions such as login, episode creation, and data validation will be tested.
- **Integration Testing:** Communication between the frontend, backend, database, and analytics modules will be tested.
- **System Testing:** The complete user workflow will be tested from login through report generation.
- **User Acceptance Testing:** Selected users will evaluate the usability and clarity of the application.
- **ML Evaluation:** If the ML module is implemented, accuracy, precision, recall, F1-score, and confusion matrix will be used for evaluation.

6 Technology Stack

The proposed technology stack is:

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, Flask, REST API
Database	MySQL
Data Processing	Python, Pandas, NumPy
Machine Learning	Scikit-learn
Visualization	Chart.js / Matplotlib
Testing	Python testing framework
Version Control	Git and GitHub
Development Environment	VS Code / PyCharm

7 Timeline

The project will be developed over the semester using an iterative development approach.

Week	Task	Deliverable
1–2	Requirement analysis and problem study	SRS and requirement specification
3–4	System architecture and database design	Architecture, ER diagram, UML diagrams
5–6	Frontend and authentication	Initial working interface
7–8	Episode recording and database integration	Core monitoring module
9–10	Dashboard and data analytics	Charts, statistics, and trend analysis
11	Treatment/session tracking and reports	Report generation module
12	Machine learning investigation and integration	Severity prediction prototype
13	Integration and system testing	Test results and bug fixes
14	User acceptance testing and refinement	Validated prototype
15	Documentation and final presentation	Final report and presentation

A buffer period will be maintained during the final development and testing stages to accommodate unexpected technical issues.

8 Resources / Budget

8.1 Human Resources

The project will be developed by two students:

- **Aditya Rajput (AIML):** Backend development, database integration, machine learning, and system integration.

- **Prabhkeerat Kaur (AIML):** Frontend development, UI/UX, documentation, testing, and analytics integration.

Both members will participate in requirements analysis, system design, testing, and final documentation.

8.2 Software Resources

The project will primarily use freely available or open-source software:

- Python.
- Flask.
- MySQL.
- Pandas and NumPy.
- Scikit-learn.
- HTML, CSS, and JavaScript.
- Git and GitHub.
- VS Code / PyCharm.

8.3 Budget

No external funding is required for the initial implementation. The software tools and development environments required for the project are freely available or accessible through the university.

Future cloud hosting or external services, if required, may introduce additional costs.

9 Risks and Mitigation

- **Limited data availability:** Initial development may use an appropriate public or synthetic dataset for the ML prototype. Future improvements can use additional anonymized data subject to appropriate consent and privacy requirements.
- **Data privacy:** Only necessary information will be collected, and user data will be protected through authentication and appropriate database security practices.
- **Medical interpretation:** The application will not claim to diagnose or treat hyperhidrosis. It will focus on monitoring, organization, and analysis of user-recorded data.
- **ML reliability:** Machine learning predictions will be treated as supplementary insights rather than medical advice.
- **Development delays:** The project will use an iterative development strategy, prioritizing the core monitoring functionality before advanced ML features.
- **User adoption:** The episode recording process will be designed to require minimal effort from the user.

10 Expected Outcomes

At the end of the project, the expected outcome is a functional web application that enables users to:

- Maintain a structured record of sweating episodes.
- View their historical sweating data.
- Understand frequency and severity trends.
- Identify commonly recorded possible triggers.
- Track treatment or session history.
- Generate an organized monitoring report.
- Receive optional ML-based personalized severity insights.

From a Software Engineering perspective, the project will demonstrate the application of requirements engineering, system architecture, database management, modular development, testing, and iterative software development.

11 Summary

HyperTrack is proposed as a software-based monitoring and management system for individuals experiencing hyperhidrosis-related excessive sweating.

The system addresses the lack of structured symptom monitoring by allowing users to record sweating episodes and associated contextual information. The collected data will be stored, processed, visualized, and converted into organized reports.

The project is primarily a Software Engineering system, with an AIML component introduced for personalized data analysis and severity prediction. The development process will emphasize requirements, architecture, modular implementation, testing, usability, privacy, and scalability.

The project therefore combines a real-world healthcare-related problem with practical Software Engineering and Artificial Intelligence and Machine Learning techniques.

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